

Toward Principled Threshold Selection for Blink Region Detection in Biological Signals: A Cross-Dataset Benchmark with LLM-Guided Candidate Derivation

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- **ABSTRACT** This study evaluated an epoch-aware pipeline for EEG blink-region detection that avoided estimating detection thresholds from concatenated blink-containing and blink-free epochs. The three-stage pipeline first screened non-overlapping epochs by peak-to-peak amplitude using an autoreject-inspired cross-validation criterion, then estimated a sample-level threshold from the suspicious-epoch pool using either a mean-based or median/MAD-based estimator with a fallback for small pools, and finally extracted blink regions and mapped them to epoch-local coordinates. BLINKER-concat, MNE-annot, Proposed-Mean, and Proposed-Med were compared on the Raja and Cao2018 naturalistic driving-EEG corpora using event-level overlap matching and macro-averaged F_1 . At 30 s, Proposed-Med achieved a pooled macro- F_1 of 0.8403, significantly exceeding BLINKER-concat at 0.7256 and MNE-annot at 0.6086. Proposed-Mean and Proposed-Med were statistically indistinguishable ($p = 1$), which indicated that the gain came from epoch screening rather than the choice of Stage-B estimator. The channel-selection ablation further showed that a single frontopolar electrode was sufficient for the strongest single-channel detection on both corpora. Finally, pooled macro- F_1 varied by only 0.0056 across epoch durations from 10 s to 120 s, and no tested duration differed from the 30 s reference after Bonferroni correction.
- **INDEX TERMS** blink detection, electroencephalography, epoch-aware detection, naturalistic driving, threshold selection

I. INTRODUCTION

Blinking is a physiological process that involves the closure and reopening of the eyelids [1]. Its primary functions are to protect the ocular surface, maintain corneal moisture, and support visual comfort [1]. Beyond these physiological functions, blink behaviour has been increasingly recognised as an important indicator in fatigue-driving research [2, 3]. Variations in blink-related features, such as blink frequency, blink duration, eye-closure duration, and blink patterns, can reflect changes in alertness and cognitive state [2, 4]. As fatigue increases, drivers may exhibit slower or prolonged blinks, which can indicate reduced vigilance and a higher risk of impaired driving performance [2]. Therefore, blink behaviour provides a useful and non-invasive marker for monitoring driver fatigue [3, 5].

In the neuroscience study of fatigue driving, blink activity can be captured through external methods, such as camera-based eye monitoring, or through electrophysiological recordings, particularly electroencephalography and electrooculography [3, 5]. Although EEG is mainly used to mea-

sure brain activity, blink-related components are often present in EEG signals because eyelid movements and eye movements generate strong electrical potentials [6, 7]. These ocular activities are especially prominent in frontal EEG electrodes due to their close proximity to the eyes [6, 8]. As a result, blink information can be extracted from EEG recordings, either by analysing the ocular components embedded in the EEG signal or by combining EEG with EOG-based measurements [5, 9]. This makes EEG/EOG recordings valuable not only for studying neural activity during fatigue but also for identifying blink-related markers associated with drowsiness [2, 10].

It is important to distinguish between two major perspectives on blink activity in the existing literature [3, 6]. In many EEG studies, blink-related activity is treated as an artifact because it contaminates neural signals and may interfere with the interpretation of brain activity [6, 7]. In such cases, blink components are commonly removed during EEG preprocessing [6, 11]. However, in fatigue-driving studies, blink activity can also be treated as meaningful information rather than

noise [3, 5]. From this perspective, blink events are identified, extracted, and analysed as potential indicators of driver fatigue [5, 10]. Blink detection methods can generally be classified into machine-learning-based, morphology-based, and thresholding-based approaches [9, 12]. Machine learning methods rely on trained models to classify blink events, while morphology-based methods identify blinks based on their characteristic waveform shape [13, 14]. Thresholding-based methods, in contrast, detect candidate blink events when the signal amplitude exceeds a certain threshold value, as blink activity typically produces large-amplitude deflections, especially in frontal channels [8, 15].

Thresholding is one of the most straightforward and widely used approaches for detecting blink activity from EEG or EOG signals [15, 16]. The simplicity of thresholding makes it computationally efficient and easy to implement, particularly in real-time fatigue-detection systems [16, 17]. Existing approaches range from single- or few-channel detectors that apply amplitude thresholds with rule-based decision logic to hybrid pipelines that combine thresholding as an initial detection step with additional candidate-refinement mechanisms [8, 9, 12, 14–16, 18–24]. Despite these developments, selecting an appropriate threshold remains challenging because blink amplitude varies across participants, electrode locations, recording devices, signal quality, and experimental conditions [20, 24]. Blink amplitude can vary across participants, electrode locations, recording devices, signal quality, and experimental conditions [20, 24]. A threshold that is too low may produce false detections due to noise or other artifacts, whereas a threshold that is too high may miss genuine blink events [16, 17]. This backdrop explains why simple one-value global thresholds can be attractive, yet brittle. Therefore, adaptive and robust threshold selection remains a critical consideration for reliable blink detection, particularly in fatigue-driving applications [5, 10].

In many fatigue-driving studies, EEG recordings are segmented into epochs that correspond to specific task events or analysis windows [25]. This epoch-structured setting introduces additional challenges for threshold-based blink detection [16, 17]. When thresholds are estimated from the entire recording, they may be biased by the presence of clean epochs that contain few or no blinks, which may distort threshold estimates and suppress blink detection in epochs where blinks are more prevalent [11, 24]. As a result, naive application of thresholding in an epoch-structured context may risk under-detection of blinks, which undermines the reliability of fatigue assessments based on blink metrics [5, 10]. Addressing this challenge requires developing blink detection methods that explicitly account for the epoch structure and can adapt threshold estimation to the specific characteristics of each epoch or subset of epochs, rather than relying on a global threshold derived from the entire recording [9, 24].

This paper makes several contributions to blink detection in fatigue driving research. First, we formalise blink detection in an epoch structured EEG setting and demonstrate the limitations of naive thresholding approaches that do not

account for the underlying epoch structure. Second, we propose a three stage blink detection pipeline comprising epoch screening, sample level threshold estimation, and blink region extraction. To our knowledge, this is among the first frameworks to integrate these three stages explicitly to address the challenges associated with epoch structured EEG recordings. Third, we introduce a robust threshold estimator based on the median and median absolute deviation (MAD) and systematically compare its performance with a mean based alternative, Proposed Mean, and conventional baseline methods, including BLINKER concat [8] and MNE annot. Finally, we provide a comprehensive benchmark of the competing methods across 30, 40, and 60 s epoch durations using two naturalistic driving EEG datasets, consisting of one dataset collected in house and one publicly available dataset [25, 26]. To support reproducibility and facilitate further research, we also developed an automated pipeline for extracting ocular indices from EEG recordings and made it freely available as a Python toolbox called *pyblinker* (TODO). The toolbox is intended to facilitate reproducible blink analysis and encourage more consistent reporting of ocular indices in EEG based fatigue research.

II. METHOD

a: Processing pipeline overview.

The proposed detector is an epoch-aware, threshold-based pipeline that transforms a continuous multichannel EEG recording into a non-overlapping epoch through a fixed sequence of five operations. The Figure 1 summarises this sequence as a flow chart, from the preprocessing, epoching, and the double thresholding which is the proposed method, through three detection stages (A, B, and C).

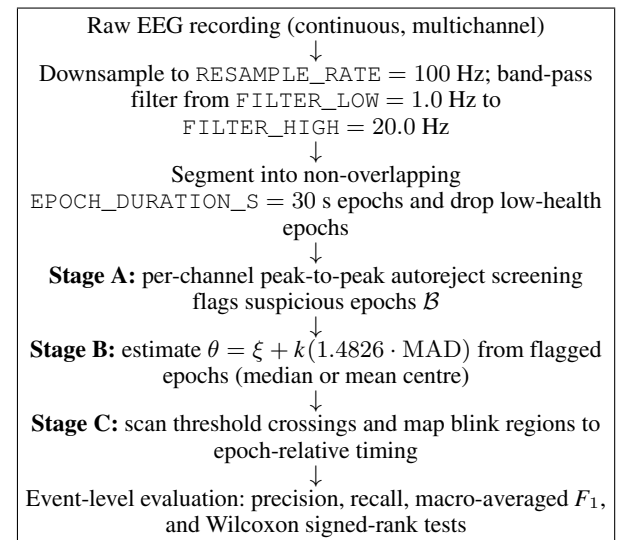


FIGURE 1. Per-epoch blink-detection pipeline. The continuous recording is downsampled and band-pass filtered, segmented into fixed-length epochs with low-health epochs removed, and then passed through the three detection stages: Stage A screens suspicious (blink-heavy) epochs by peak-to-peak amplitude, Stage B estimates a robust sample-level amplitude threshold from those epochs, and Stage C localises blink regions by scanning threshold crossings. Detections are finally scored against the ground truth at the event level.

b: Step-by-step description.

First, the continuous recording is downsampled to a common sampling grid of 100 Hz (RESAMPLE_RATE). Standardising the sampling rate makes the amplitude thresholds and the minimum-duration gate comparable across recordings acquired on different hardware, and reduces the per-sample computational cost of the downstream stages without discarding any information needed to resolve a blink, whose spectral content lies well below the new Nyquist frequency. Second, a zero-phase band-pass filter from 1.0 Hz (FILTER_LOW) to 20.0 Hz (FILTER_HIGH) is applied; this legacy BLINKER-style band removes slow drifts and DC offsets at the low end and high-frequency muscle and line noise at the high end, isolating the low-frequency, high-amplitude deflection that characterises an eye blink. Third, each filtered recording is split into non-overlapping, fixed-length epochs of 30 s (EPOCH_DURATION_S); epochs whose health metric falls below the acceptance criterion are dropped so that grossly corrupted segments do not distort the data-driven thresholds estimated later. Epoching is what makes the thresholds *adaptive*: rather than deriving a single level for the entire session, the pipeline treats each short epoch as a local statistical context.

In Stage A, a per-channel peak-to-peak (PTP) amplitude statistic is computed for every epoch, and an *autoreject*-based procedure estimates a channel-specific PTP threshold from the data. An aggregation rule over a chosen channel-selection group, by default flagging an epoch whenever *any* channel in the group exceeds its threshold, marks the epoch as “suspicious” (blink-heavy). The output of Stage A is therefore a set $\mathcal{B}$ of epochs that are enriched for blink-related activity, which restricts the subsequent amplitude analysis to the segments where blinks are most likely to occur. In Stage B, a robust sample-level amplitude threshold is estimated using only the samples pooled from the flagged epochs, as $\theta = \xi + k(1.4826 \cdot \text{MAD})$, where the centre ξ is the *median* in the primary configuration (or the *mean* in the contrast configuration), MAD is the median absolute deviation, the factor 1.4826 rescales the MAD to a standard-deviation-consistent dispersion, and k is the sensitivity multiplier. Estimating the threshold from the flagged epochs concentrates the calculation on blink-enriched signal, while the median-plus-MAD construction keeps the estimate stable in the presence of isolated extreme samples. If too few epochs are flagged, the estimate falls back to the full valid-epoch set so that the threshold is never derived from an unreliably small sample. In Stage C, the threshold is scanned across the signal in the manner of the Kleifges crossing detector: maximal contiguous runs of supra-threshold samples define candidate blink regions, brief crossings shorter than a minimum-duration gate are discarded, and the accepted regions are mapped from their concatenated coordinates back to epoch-relative onset and offset times. Finally, the detected regions are compared with the ground-truth annotations at the *event* level using overlap-based matching, yielding precision, recall, and F_1 ; these are

macro-averaged across sessions and compared between configurations with Wilcoxon signed-rank tests, as detailed in Section II-E. Recordings were segmented into fixed-length epochs, with 30 s used as the primary epoch length and 10, 20, 40, and 60 s additionally evaluated in an epoch-sensitivity sweep. The proposed detector was implemented as a three-stage epoch-aware threshold pipeline. First, suspicious/bad epochs were screened using an autoreject-based procedure with a fixed random state and a requirement that at least one epoch be flagged. Second, a sample-level amplitude threshold was estimated from the screened epochs using a robust centre and a standard-deviation multiplier of 3.5. Third, threshold crossings were expanded into blink regions. The detailed equations for each stage were given in the following subsections rather than duplicated in this overview.

A. SELECTION OF BAD EPOCHS

The first stage of the proposed pipeline performs epoch-level screening using an autoreject-inspired threshold selection procedure. Its purpose is not to estimate the onset or offset of a blink, but to identify epochs whose amplitudes are sufficiently abnormal to warrant subsequent sample-level localization. Let $X \in \mathbb{R}^{N \times C \times T}$ denote the epoched multichannel signal after preprocessing, where N is the number of valid epochs, C is the number of channels, and T is the number of samples per epoch. The sample at epoch $i \in \mathcal{E} = \{1, \dots, N\}$, channel $c \in \mathcal{C} = \{1, \dots, C\}$, and within-epoch time index $t \in \{1, \dots, T\}$ is denoted by $x_{i,c,t}$.

For each epoch and channel, an amplitude-range statistic is computed using the peak-to-peak value

$$a_{i,c} = \max_{1 \leq t \leq T} x_{i,c,t} - \min_{1 \leq t \leq T} x_{i,c,t}.$$

Large values of $a_{i,c}$ indicate that the corresponding epoch contains a high-amplitude excursion in channel c , a pattern consistent with ocular or other transient artifacts. Rather than imposing a fixed rejection level, the pipeline estimates a channel-specific threshold τ_c^* from the data. Candidate thresholds are taken from the empirical range of peak-to-peak amplitudes for that channel, denoted $\mathcal{Q}_c = \{a_{i,c} : i \in \mathcal{E}\}$, and are evaluated by a cross-validation criterion.

Specifically, let $\{(\mathcal{T}_k, \mathcal{V}_k)\}_{k=1}^K$ be cross-validation splits of the epoch indices into training sets $\mathcal{T}_k$ and validation sets $\mathcal{V}_k$. For a candidate threshold $\tau \in \mathcal{Q}_c$, the training epochs retained for channel c in fold k are

$$\mathcal{G}_{k,c}(\tau) = \{i \in \mathcal{T}_k : a_{i,c} \leq \tau\}.$$

These retained epochs define a threshold-dependent training template

$$\bar{x}_{k,c}^{(\tau)}(t) = \frac{1}{|\mathcal{G}_{k,c}(\tau)|} \sum_{i \in \mathcal{G}_{k,c}(\tau)} x_{i,c,t},$$

where candidates yielding an empty retained set are treated as inadmissible. The corresponding validation reference is computed as the pointwise median across the validation epochs,

$$\tilde{x}_{k,c}(t) = \text{median}_{i \in \mathcal{V}_k} x_{i,c,t}.$$

The use of a median validation reference makes the criterion less sensitive to large validation-set artifacts. The validation loss for channel c is then defined as

$$\mathcal{J}_c(\tau) = \frac{1}{K} \sum_{k=1}^K \left[\frac{1}{T} \sum_{t=1}^T \left\{ \bar{x}_{k,c}^{(\tau)}(t) - \tilde{x}_{k,c}(t) \right\}^2 \right]^{1/2}.$$

The selected channel threshold is the candidate that minimizes the average validation discrepancy,

$$\tau_c^* = \arg \min_{\tau \in \mathcal{Q}_c} \mathcal{J}_c(\tau).$$

In practice, this objective may be searched by sequential Bayesian optimization or by a random search over candidate peak-to-peak thresholds; both choices retain the same validation objective and rejection rule.

After the thresholds $\{\tau_c^*\}_{c=1}^C$ have been estimated, each epoch receives a channel-level artifact indicator

$$z_{i,c} = \mathbf{1}\{a_{i,c} > \tau_c^*\},$$

where $\mathbf{1}\{\cdot\}$ is the indicator function. The present pipeline uses the union of the channel-level decisions to form the epoch-level screening mask,

$$b_i = \mathbf{1} \left\{ \sum_{c=1}^C z_{i,c} \geq 1 \right\} = \mathbf{1} \{ \exists c \in \mathcal{C} : a_{i,c} > \tau_c^* \}.$$

The set of suspicious epochs is therefore

$$\mathcal{B} = \{i \in \mathcal{E} : b_i = 1\}.$$

These epochs are interpreted as likely blink-contaminated or artifact-heavy segments. They provide the restricted data subset from which the next stage estimates sample-level blink-region thresholds.

B. COMPUTATION OF SAMPLE-LEVEL BLINK-REGION THRESHOLD CANDIDATES

The second stage converts the epoch-level artifact decision into a sample-level thresholding problem. The autoreject-style screening in the previous stage identifies a set of suspicious epochs $\mathcal{B}$, but its thresholds τ_c^* are defined on peak-to-peak amplitudes over whole epochs. Such thresholds are therefore appropriate for deciding whether an epoch is abnormal, but they are not directly interpretable as sample-wise blink crossing levels. Stage B estimates a separate channel-specific threshold in the amplitude domain of the time series itself.

Let $\mathcal{S}$ denote the set of epochs used for estimating the sample-level threshold. In the primary analysis, $\mathcal{S} = \mathcal{B}$, so that the estimate is driven by epochs already identified as likely to contain blink-related activity. If too few suspicious epochs are available, the method falls back to the full valid epoch set $\mathcal{E}$ to avoid estimating a threshold from an insufficient sample:

$$\mathcal{S} = \begin{cases} \mathcal{B}, & |\mathcal{B}| \geq n_{\min}, \\ \mathcal{E}, & |\mathcal{B}| < n_{\min}, \end{cases}$$

where $n_{\min}$ is the minimum number of screened epochs required for a stable estimate.

For each channel c , all samples from the selected epochs are pooled into a one-dimensional empirical sample set

$$\mathcal{Y}_c = \{x_{i,c,t} : i \in \mathcal{S}, 1 \leq t \leq T\}.$$

This pooling step marks the methodological transition from epoch-level rejection to sample-level localization: the quantities used in Stage B are no longer peak-to-peak ranges $a_{i,c}$, but the individual signal amplitudes $x_{i,c,t}$ that will later be thresholded to form candidate blink regions.

The sample-level threshold is estimated using a robust location-plus-dispersion form. Let ξ_c be the channel-specific center of the sample distribution. The default choice is the median,

$$\xi_c = \text{median}_{y \in \mathcal{Y}_c} y,$$

although the arithmetic mean may be substituted when a mean-centered threshold is required for comparison. Dispersion is estimated by the median absolute deviation (MAD),

$$\text{MAD}_c = \text{median}_{y \in \mathcal{Y}_c} |y - \text{median}_{u \in \mathcal{Y}_c} u|,$$

and is rescaled to a standard-deviation-compatible quantity by

$$\sigma_c^{\text{rob}} = 1.4826 \text{ MAD}_c.$$

The constant 1.4826 is the usual normal-consistency factor for the MAD. For a sensitivity parameter $k > 0$, the resulting sample-level blink-region threshold candidate for channel c is

$$\theta_c = \xi_c + k \sigma_c^{\text{rob}}.$$

The set

$$\Theta = \{\theta_c : c \in \mathcal{C}\}$$

constitutes the channel-wise threshold candidates passed to the blink-region extraction stage.

This construction deliberately separates the statistical role of the two thresholds in the pipeline. The quantities τ_c^* from Stage A are epoch-rejection thresholds defined on peak-to-peak amplitudes, whereas θ_c is a sample-level amplitude threshold defined on the time-series samples themselves. Estimating θ_c from $\mathcal{S}$ concentrates the calculation on signal segments enriched for blink-like excursions, while the robust center and MAD-scaled dispersion reduce the influence of isolated extreme peaks within those same segments.

C. EXTRACTION OF BLINK REGIONS

The final stage applies the sample-level thresholds $\Theta = \{\theta_c\}_{c=1}^C$ to obtain channel-wise blink regions. For each channel c , the epochs selected for localization are ordered as $\mathcal{S} = \{i_1, \dots, i_R\}$, where $R = |\mathcal{S}|$. Their samples are concatenated into a one-dimensional signal

$$u_c[n] = x_{i_r,c,t}, \quad n = (r-1)T + t, \quad r \in \{1, \dots, R\}, \quad t \in \{1, \dots, T\}$$

Thus $u_c \in \mathbb{R}^{RT}$ is the channel-specific signal on which blink onsets and offsets are estimated. If the analysis is run in the

fallback mode described above, $\mathcal{S}$ is the full valid epoch set; otherwise it is the set of epochs identified as suspicious in Stage A.

Candidate blink samples are defined by a threshold-crossing indicator

$$h_c[n] = 1\{u_c[n] > \theta_c\}, \quad n = 1, \dots, RT.$$

After any required polarity normalization, $h_c[n] = 1$ indicates that the sample amplitude exceeds the channel-specific blink-region threshold. Blink regions are then defined as maximal contiguous runs of supra-threshold samples. A candidate interval for channel c is written as the half-open index range $[s_{c,r}, e_{c,r})$, where

$$h_c[n] = 1 \quad \text{for all } n = s_{c,r}, \dots, e_{c,r} - 1,$$

with either $s_{c,r} = 1$ or $h_c[s_{c,r} - 1] = 0$, and either $e_{c,r} = RT + 1$ or $h_c[e_{c,r}] = 0$. Equivalently, the onset $s_{c,r}$ is the first sample in a run that crosses above θ_c , and the offset $e_{c,r}$ is the first subsequent sample at which the signal has returned to the sub-threshold state.

To suppress brief threshold crossings that are unlikely to represent complete blinks, each candidate interval is required to exceed a minimum duration. Let f_s be the sampling frequency and let $\ell_{\min}$ be the minimum blink duration in seconds. The corresponding sample length is

$$L_{\min} = \ell_{\min} f_s.$$

The accepted blink-region set for channel c is

$$\mathcal{R}_c = \{(s_{c,r}, e_{c,r}) : e_{c,r} - s_{c,r} > L_{\min}\}.$$

This duration gate is applied independently for each channel. Consecutive supra-threshold samples are merged into the same region by construction, while two runs separated by at least one sub-threshold sample are treated as distinct candidate blink regions.

Finally, the accepted intervals are mapped from concatenated coordinates back to epoch-local coordinates using the known epoch boundaries. If an accepted onset $s_{c,r}$ lies in the concatenated segment corresponding to epoch i_q , its within-epoch onset sample is

$$\hat{s}_{c,r} = s_{c,r} - (q - 1)T,$$

and its onset time is $(\hat{s}_{c,r} - 1)/f_s$ seconds relative to the beginning of that epoch. The interval duration is

$$\hat{d}_{c,r} = \frac{e_{c,r} - s_{c,r}}{f_s}.$$

The final output of this stage is therefore a channel-indexed collection of blink regions, each represented by an epoch identifier, a channel label, an onset index or time, and an offset or duration. The extraction remains channel-wise throughout: the threshold θ_c estimated for channel c is applied only to that channel's signal, and the resulting regions $\mathcal{R}_c$ are retained as channel-specific blink candidates.

D. EPOCH SEGMENTATION AND DETECTION PIPELINE

E. PROBLEM DEFINITION

Let a blink annotation be an interval $g_i = [s_i, e_i]$ and a detector output be an interval $d_j = [\hat{s}_j, \hat{e}_j]$. A detected region is counted as a true positive if it overlaps at least one unmatched ground-truth region. Unmatched detections are false positives; unmatched ground-truth regions are false negatives. This event-level treatment follows the logic of interval-based detection rather than pointwise accuracy, which is appropriate because blink boundaries are uncertain, annotation granularity varies, and non-event samples dominate the timeline [27, 28].

F. DATASETS

Two naturalistic driving-EEG corpora were evaluated: Raja and Cao2018. The Raja corpus comprised 46 complete sessions recorded with a 128-channel EGI system (ANT Neuro), referenced to CPz under a modified international 10-20 layout, during a simulated lane-keeping driving task; ground-truth blink annotations were derived from human EAR/EOG labelling. The Cao2018 corpus was the sustained-attention driving corpus reported by [25], in which 32-channel EEG was recorded during an approximately 90-minute lane-keeping task; 58 complete sessions were used, and 4 non-complete sessions were excluded. Because the published preprocessed version of Cao2018 removed eye blinks, the raw recordings were used and paired with external blink-region annotations.

G. PREPROCESSING

All detectors used identical preprocessing, consisting of a 1–20 Hz band-pass filter, no resampling, and EEG-channel selection. The channel scope was dataset specific: for Raja, the analysis was restricted to a minimal frontal montage comprising three channels (E3, E9, and E22), whereas for Cao2018, all 32 EEG channels were provided to the pipeline and the best channel was selected separately for each session. Thus, Raja channel selection was confined to frontal sites by construction, while Cao2018 channel selection was performed over the full montage.

H. DETECTOR CONFIGURATIONS

Four detector configurations were compared under identical preprocessing and epoching. The proposed methods were Proposed-Med, which implemented the three-stage pipeline with a median centre, and Proposed-Mean, which used a mean centre. The conventional baselines were BLINKER-concat and MNE-annot.

I. EXPERIMENT OVERVIEW

The experiments follow a deliberate logical progression. Experiment 1 first selects the best Stage-A channel-selection configuration; Experiment 2 then benchmarks that chosen configuration against competing detectors; and Experiment 3 stress-tests the resulting pipeline against the one analysis choice most likely to be imposed by the experimental

paradigm rather than by the detector, namely epoch length. Fixing the channel configuration first, before any baseline comparison, ensures that the headline comparison in Experiment 2 uses a principled spatial selection rather than an arbitrary default, and that the robustness check operates on the same ideal configuration. Each experiment is summarised below in terms of its purpose, motivation, and objective.

a: Experiment 1 – Channel-selection strategies.

Purpose: determine which Stage-A channel-selection group yields the best detector, evaluated on both datasets (Raja and Cao2018) and under both estimators (median and mean), before any other experiment is run. **Motivation:** the current default rule flags an epoch whenever *any* channel over *all* channels exceeds its peak-to-peak threshold, which can be triggered by a single noisy non-ocular channel and inflate the number of flagged epochs; a more selective spatial group may screen blinks more cleanly. **Objective:** run the complete Stage A→B→C pipeline on each candidate channel group and select the best group by weighing multiple criteria rather than a single metric. The decision criteria are:

- (i) Downstream event-level F_1 of the full pipeline when restricted to the group; this is the ultimate quantity the detector is optimised for and is therefore the primary criterion.
- (ii) Stage-A epoch-level F_1 of the suspicious-epoch selection itself, scored against epoch-level ground truth (an epoch is positive if it contains at least one annotated blink); this isolates the quality of the screening stage from the downstream localisation.
- (iii) Number of channels for Stage-A requires, as a measure of montage cost, simplicity, and portability; a group that performs comparably with fewer channels is preferred because it is cheaper to acquire and easier to deploy.
- (iv) False-positive ratio per epoch flagged. The proportion of epochs flagged and the overall proportion of epochs flagged; a low value indicates robustness to artefacts and avoids over-flagging that would dilute the Stage-B threshold estimate.
- (v) Cross-dataset consistency of the chosen group's behaviour between Raja and Cao2018; a group that is best on one dataset but poor on the other does not generalise.
- (vi) Best channel stability. Which channel (or channels) drives detection across sessions and subjects; an unstable best channel signals an overfitted, fragile selection.
- (vii) Estimator consistency of the group's ranking under the median versus mean centre at Stage B; a group whose advantage survives both estimators is more trustworthy than one that depends on a particular centre.

The configuration that best balances these criteria is designated the ideal configuration and is carried forward unchanged into Experiments 2 and 3.

b: Experiment 2 – Strategy comparison.

Purpose: benchmark the proposed pipeline, configured with the ideal channel group from Experiment 1, against competing detection algorithms. **Motivation:** a method is only convincing if it outperforms the established alternatives under identical preprocessing, epoching, and scoring. **Objective:** compare BLINKER-concat, MNE-annot, Proposed-Mean, and Proposed-Med at the event level on both datasets, reporting precision, recall, and F_1 with paired significance tests, and isolating both the estimator contrast (median versus mean) and the cross-dataset generalisation gap.

c: Experiment 3 – Epoch-duration stability.

Purpose: characterise how sensitive the proposed pipeline is to the choice of epoch length. **Motivation:** the 30 s primary epoch is a design choice, and a robust method should not depend critically on it. **Objective:** sweep the epoch duration (10/20/40/50/60/120 s around the 30 s reference) and quantify the resulting variation in event-level F_1 , testing each duration against the reference with Wilcoxon signed-rank tests.

J. EVALUATION AND STATISTICAL ANALYSIS

Detection was scored at the event level: a detected interval that overlapped an unmatched ground-truth interval was counted as a true positive, unmatched detections were counted as false positives, and unmatched ground-truth intervals were counted as false negatives. Event-level precision, recall, and F_1 were macro-averaged across sessions, and event-level F_1 was preferred over accuracy because blink events were sparse relative to non-blink samples. A detection was matched to a ground-truth interval at an intersection-over-union tolerance of 0.1. Methods were compared using paired two-tailed Wilcoxon signed-rank tests on session-level F_1 , with Bonferroni correction applied to the six comparisons among the four methods and to the six non-reference durations of the epoch sweep against the 30 s reference; rank-biserial effect sizes were also reported.

III. EXPERIMENTS AND RESULTS

A. EXPERIMENTAL SETUP

All experiments are evaluated on the driving corpora described in Section II-F. Unless otherwise stated, each continuous session is divided into non-overlapping 30-second epochs, and the evaluation is conducted at the event level using the overlap-based matching criterion introduced in Section II-E.

a: Conditions.

Four conditions are compared throughout.

BLINKER-Blink algorithm is applied to the full concatenation of all valid epochs, treating the session as a single continuous signal and deriving a single threshold from the entire recording [8]. This condition represents the current practice of running a

continuous-signal detector on epoch-structured data without any epoch-aware adaptation.

HotMNE-Python `annotate_amplitude` routine, which marks contiguous segments whose peak-to-peak amplitude exceeds a configurable threshold, used here with its default parameterisation. This baseline represents widespread preprocessing practice in the neuroimaging community and provides a reference point grounded in community tooling.

Proposed Multi three-stage pipeline described in Section II, with the sample-level threshold at Stage B estimated from the arithmetic mean and standard deviation of samples pooled from suspicious epochs.

Proposed Med three-stage pipeline with the sample-level threshold at Stage B estimated from the median and median absolute deviation (MAD). This is the primary proposed configuration.

b: Aggregation.

Every condition is summarised at its *best-channel-per-session* operating point: for each session the channel or channel group with the highest event-level F_1 is selected, and those per-session values are then averaged. The same rule is applied to all four conditions, so no condition is given a selection advantage the others do not receive.

B. EVALUATION METRICS

a: Event-level precision, recall, and F_1 .

A detected interval is counted as a true positive if it overlaps at least one unmatched ground-truth blink region; unmatched detections are false positives, and unmatched ground-truth regions are false negatives. Per-session precision, recall, and F_1 are computed from these event-level counts.

b: Macro versus micro F_1 .

We adopt **macro-averaged F_1** as the primary reporting metric. Macro F_1 computes precision, recall, and F_1 independently for each session and then reports the unweighted mean of the session-level F_1 scores. This gives every session equal weight regardless of how many blink events it contains. The goal of epoch-based blink detection in driving studies is to produce a detector that works reliably session by session, not one that merely accumulates favourable aggregate counts. Micro F_1 , which pools all true positives, false positives, and false negatives across sessions before computing a single ratio, is dominated by sessions with large blink counts and can mask consistent failure modes on sessions with atypically few blinks.

c: Wilcoxon signed-rank test.

Pairwise comparisons between conditions use the Wilcoxon signed-rank test on matched pairs of session-level F_1 scores. The test is preferred over a paired t -test because session-level F_1 distributions are bounded and often skewed. For

comparisons in which the proposed method is hypothesised to outperform a baseline, a one-tailed test is used; stability analyses use two-tailed tests. Bonferroni correction is applied for the number of conditions compared within each experiment, and we report exact p -values alongside the matched-pairs rank-biserial correlation r as an effect-size estimate.

C. EXPERIMENT 1: EEG CHANNEL-SELECTION ABLATION

a: Motivation.

EEG blink artefacts are largest over frontal scalp sites, but it is unclear how much performance degrades when only a subset of the available channels is used. This experiment ablates channel groups and individual electrodes to characterise the performance-versus-coverage trade-off and to identify a compact, reliable channel set.

b: Design.

The full pipeline (30-second epochs, median centre) was applied to every session of both datasets and evaluated *channel by channel*: each electrode's macro-averaged precision, recall and F_1 were read directly from the per-channel rows of the Experiment 1 results (the `selection=all_channel` rows), averaged over sessions, with no region-level pooling. For the Raja EGI-128 montage the native electrode labels are accompanied by their 10–20 equivalents through the channel map of Table 1; the Cao2018 montage is already in 10–20 nomenclature.

The channel-by-channel ablation showed that detection performance was concentrated specifically at the frontopolar electrodes rather than across the frontal region as a whole (Table 2). In Raja, Fp2 (EGI E9) achieved $F_1 = 0.868$ and Fp1 (EGI E22) achieved $F_1 = 0.863$, whereas performance decreased at AF4 and AF3 to $F_1 = 0.803$ and $F_1 = 0.738$, respectively, and declined further at F3 and F4 to $F_1 = 0.456$ and $F_1 = 0.452$. The same localisation was observed in Cao2018, where Fp1 and Fp2 achieved $F_1 = 0.797$ and $F_1 = 0.784$, compared with $F_1 = 0.595$ at F3 and $F_1 = 0.580$ at F4. Performance collapsed outside frontal sites, with the worst electrodes reaching $F_1 = 0.017$ in Raja and $F_1 = 0.028$ in Cao2018. Despite the different montages, both corpora therefore localised the optimum to the corresponding Fp1/Fp2 frontopolar sites, with the Raja EGI equivalents listed in Table 1. The best single frontopolar electrode approached the whole-cap oracle values of $F_1 = 0.8894$ and $F_1 = 0.8175$, indicating that a dense montage was not required.

1) Regional Profile and Single-Channel Operation

Consistent with the channel-level ablation, region-level performance was concentrated at the frontopolar pair rather than across the frontal region as a whole (Table 3; Figure 2). On Raja, the two frontopolar electrodes achieved precision, recall, and F_1 of 0.8636, 0.8934, and 0.8657, respectively, compared with 0.7564, 0.5676, and 0.6124 across the four remaining frontal electrodes. On Cao2018, the corresponding frontopolar values were 0.7389, 0.8924, and 0.7906, whereas

TABLE 1. Raja EGI 128-channel (HydroCel GSN) to 10–20 scalp-location mapping, taken from the `egi_pair` block of `brain_region_raja.yaml`. Results are reported throughout by 10–20 location; this table gives the native EGI index for each one. The region column is the assignment used by the Experiment 1 run; `midline_or_outside` marks electrodes outside the curated left/right regions. Cao2018 is acquired directly in 10–20 nomenclature and needs no mapping.

10–20	EGI	Region
F7	E33	–
T7	E40	–
O2	E84	–
T8	E109	–
F8	E122	–
FC1	E13	central_left
C3	E26	central_left
FC5	E28	central_left
C4	E104	central_right
FC2	E112	central_right
FC6	E117	central_right
Fp1	E22	frontal_left
AF3	E23	frontal_left
F3	E24	frontal_left
AF4	E3	frontal_right
Fp2	E9	frontal_right
F4	E124	frontal_right
Fz	E11	midline_or_outside
Pz	E62	midline_or_outside
Oz	E75	midline_or_outside
Cz	E129	midline_or_outside
P7	E59	occipital_left
P3	E60	occipital_left
PO3	E67	occipital_left
O1	E70	occipital_left
PO4	E77	occipital_right
P4	E85	occipital_right
P8	E91	occipital_right
CP1	E37	parietal_left
CP5	E47	parietal_left
CP2	E87	parietal_right
CP6	E98	parietal_right

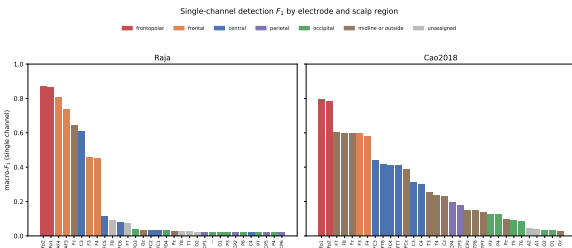


FIGURE 2. Single-channel detection F_1 for every electrode, coloured by scalp region.

the two remaining frontal electrodes achieved 0.7217, 0.5405, and 0.5875. The frontal-plus-frontopolar mean F_1 was 0.697 compared with 0.080 across 26 non-frontal electrodes on Raja, and 0.689 compared with 0.240 across 28 non-frontal electrodes on Cao2018 (Figure 3). The frontal versus non-frontal separation was therefore larger on Raja. This cross-dataset difference coincided with differences in montage composition, particularly for midline-or-outside electrodes, with $n=3$ on Raja and $n=12$ on Cao2018. The single-channel results retained the same frontopolar concentration (Figure 4).

TABLE 2. Experiment 1 channel-by-channel detection performance of Proposed-Med (median centre), organised by scalp region with *no* region-level aggregation. Each entry is the median-thresholded detector evaluated on a single channel, with macro-averaged precision, recall and F_1 averaged over all sessions (values read directly from the `selection=all_channel` rows of the Experiment 1 results). Electrodes are identified by their 10–20 scalp location on both corpora; for Raja the corresponding native EGI HydroCel index is given alongside it (full mapping in Table 1). Best F_1 per dataset in bold.

Region	Channel	EGI	Precision	Recall	F_1
<i>Raja (EGI 128, 46 sessions)</i>					
frontal_left	Fp1	E22	0.865	0.889	0.863
	AF3	E23	0.817	0.720	0.738
	F3	E24	0.642	0.397	0.456
frontal_right	Fp2	E9	0.863	0.897	0.868
	AF4	E3	0.869	0.777	0.803
	F4	E124	0.698	0.377	0.452
central_left	C3	E26	0.690	0.602	0.611
	FC5	E28	0.385	0.075	0.116
	FC1	E13	0.293	0.017	0.031
central_right	FC6	E117	0.374	0.051	0.081
	FC2	E112	0.276	0.018	0.031
	C4	E104	0.157	0.010	0.019
parietal_left	CP1	E37	0.150	0.012	0.022
	CP5	E47	0.122	0.010	0.018
	CP2	E87	0.173	0.011	0.020
parietal_right	CP6	E98	0.118	0.010	0.017
	PO3	E67	0.141	0.023	0.036
occipital_left	O1	E70	0.123	0.012	0.021
	P3	E60	0.124	0.012	0.021
	P7	E59	0.118	0.010	0.018
occipital_right	PO4	E77	0.143	0.018	0.030
	–	E83	0.117	0.013	0.021
	P8	E91	0.117	0.011	0.019
midline_or_outside	P4	E85	0.114	0.010	0.018
	Fz	E11	0.805	0.599	0.642
	Oz	E75	0.124	0.021	0.033
unassigned	Pz	E62	0.138	0.016	0.028
	F8	E122	0.296	0.060	0.092
	F7	E33	0.209	0.050	0.074
	T8	E109	0.170	0.015	0.027
	T7	E40	0.151	0.014	0.025
	O2	E84	0.117	0.013	0.022
<i>Cao2018 (10–20, 58 sessions)</i>					
frontal_left	Fp1	–	0.740	0.905	0.797
	F3	–	0.726	0.549	0.595
	F4	–	0.717	0.532	0.580
frontal_right	Fp2	–	0.738	0.879	0.784
	F4	–	0.717	0.532	0.580
central_left	FC3	–	0.668	0.373	0.441
	FT7	–	0.621	0.336	0.408
	C3	–	0.595	0.243	0.311
central_right	FT8	–	0.643	0.344	0.416
	FC4	–	0.647	0.344	0.410
	C4	–	0.579	0.234	0.302
parietal_left	CP3	–	0.482	0.127	0.178
	CP4	–	0.492	0.140	0.196
	P3	–	0.425	0.085	0.128
parietal_right	T5	–	0.354	0.060	0.093
	O1	–	0.227	0.018	0.031
	P4	–	0.429	0.083	0.127
occipital_left	T6	–	0.351	0.052	0.083
	O2	–	0.256	0.019	0.033
	F7	–	0.709	0.571	0.603
occipital_right	F8	–	0.708	0.557	0.597
	Fz	–	0.727	0.556	0.597
	FCz	–	0.633	0.323	0.389
midline_or_outside	T3	–	0.524	0.188	0.256
	T4	–	0.535	0.167	0.236
	Cz	–	0.511	0.174	0.231
unassigned	CPz	–	0.436	0.103	0.149
	TP8	–	0.448	0.098	0.149
	TP7	–	0.410	0.091	0.135
	Pz	–	0.389	0.062	0.098
	Oz	–	0.225	0.016	0.028
	A2	–	0.210	0.027	0.046
	A1	–	0.173	0.022	0.037

TABLE 3. Experiment 1 single-channel detection performance collapsed to coarse scalp regions (Proposed-Med, median centre). Each region row averages the per-channel macro precision, recall and F_1 over the electrodes it contains; the frontopolar pair (Raja E22/E9, Cao2018 FP1/FP2) is reported separately from the remaining frontal sites. n is the number of electrodes in the region.

Dataset	Region	n	Precision	Recall	F_1
Raja	frontopolar	2	0.8636	0.8934	0.8657
	frontal	4	0.7564	0.5676	0.6124
	central	6	0.3626	0.1288	0.1480
	parietal	4	0.1409	0.0110	0.0194
	occipital	8	0.1247	0.0138	0.0230
	midline_or_outside	3	0.3557	0.2121	0.2342
	unassigned	5	0.1885	0.0305	0.0480
Cao2018	frontopolar	2	0.7389	0.8924	0.7906
	frontal	2	0.7217	0.5405	0.5875
	central	6	0.6257	0.3123	0.3813
	parietal	2	0.4870	0.1332	0.1870
	occipital	6	0.3404	0.0529	0.0824
	midline_or_outside	12	0.5213	0.2422	0.2890
	unassigned	2	0.1914	0.0244	0.0410

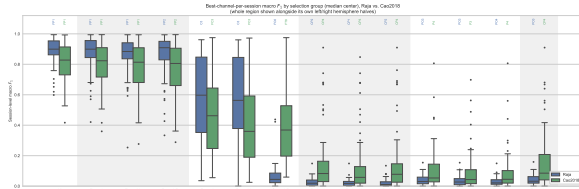


FIGURE 3. Session-level macro F_1 by channel-selection group (median centre), Raja versus Cao2018.

2) Stability of the Channel Choice

a: Design.

Experiment 1 identifies the best channel on average; this analysis asks how concentrated that choice is in practice. The per-session best channel is pooled over the four detection conditions of Experiment 2, giving both a selection frequency (Table 4, Figure 5) and an agreement rate between conditions (Table 5).

TABLE 4. Best-channel selection frequencies pooled over the four conditions (best-channel-per-session). The 5 most frequently selected electrodes are listed per dataset, with the count and the fraction of all session $\times$ condition selections.

Dataset	Summary	Frequencies
Raja	Channels	Fp2 75/184 (0.408); Fp1 67/184 (0.364); AF4 14/184 (0.076); AF3 11/184 (0.060); Fz 4/184 (0.022)
Cao2018	Channels	Fp1 119/232 (0.513); Fp2 57/232 (0.246); F8 11/232 (0.047); Fz 11/232 (0.047); F7 9/232 (0.039)

Within this frontopolar concentration, best-channel selection consistently favoured Fp1 and Fp2, although the precise choice between them varied across the four conditions (Table 4; Figure 5). In Raja, Fp2 accounted for 75/184 (0.408) selections and Fp1 for 67/184 (0.364), whereas the next most frequently selected channels were AF4 at 14/184 (0.076) and AF3 at 11/184 (0.060). In Cao2018, Fp1 accounted for 119/232 (0.513) selections and Fp2 for 57/232 (0.246), fol-

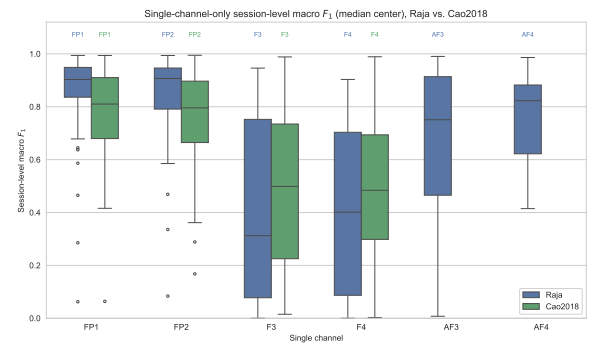


FIGURE 4. Session-level macro F_1 for the single-channel selection groups.

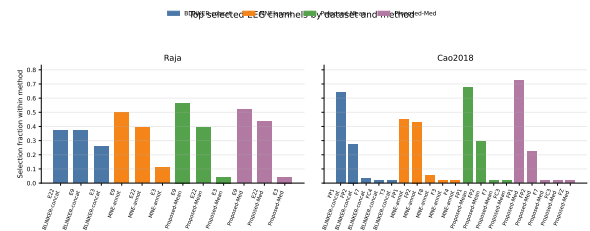


FIGURE 5. Best-channel selection frequency, pooled over the four conditions.

lowed by F8 and Fz at 11/232 (0.047) each. However, full agreement on the exact best channel occurred in only 9/46 (0.196) Raja sessions and 6/58 (0.103) Cao2018 sessions, with mean pairwise agreement of 0.489 and 0.408, respectively (Table 5). Pooled full agreement was 15/104 (0.144), and pooled mean pairwise agreement was 0.444. Thus, the selected scalp region was stable, but the precise choice between Fp1 and Fp2 was not.

D. EXPERIMENT 2: STRATEGY COMPARISON

a: Motivation.

In driving and other epoch-structured paradigms it is common to run a continuous-signal blink detector on the raw or minimally preprocessed recording and map detections back onto the epoch grid. BLINKER-concat embodies this practice: its threshold is derived from a signal pool that mixes blink-heavy and blink-free epochs. The central claim of the proposed pipeline is that this mixing dilutes the amplitude statistics used for threshold estimation, distorting threshold placement and shifting the precision–recall trade-off. MNE-annot provides a second reference point grounded in community tooling: establishing its performance under an event-level blink criterion, for which it was not designed, characterises the gap that specialised methods must close.

b: Hypothesis.

Naive concatenation will produce systematically lower session-level F_1 than the epoch-aware pipeline through an unfavourable precision–recall trade-off.

TABLE 5. Stability of the best-channel choice across the four conditions. Full agreement is the fraction of sessions where all four conditions select the same best channel; per-condition agreement is the mean fraction of the other three conditions selecting the same best channel.

Dataset	Sessions	Full agreement	Mean pairwise	BLINKER-concat	MNE-annot	Proposed-Mean	Proposed-Med
Raja	46	9/46 (0.196)	0.489	0.442	0.326	0.587	0.601
Cao2018	58	6/58 (0.103)	0.408	0.397	0.218	0.506	0.511
Pooled	104	15/104 (0.144)	0.444	0.417	0.266	0.542	0.551

c: Design.

All four conditions are applied to every session under 30-second epochs on the `all_channel` gate. Per-session macro F_1 , precision and recall are summarised in Table 6.

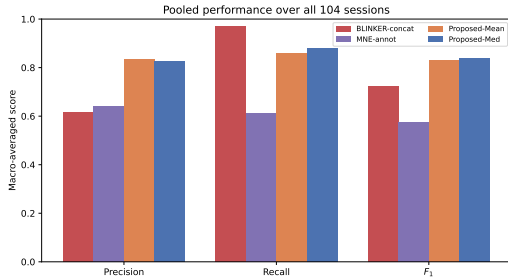


FIGURE 6. Macro-averaged precision, recall, and F_1 for the four conditions over all sessions.

In the strategy comparison, the two proposed configurations achieved higher macro- F_1 than both baselines on Raja, Cao2018, and the pooled evaluation (Table 6). Proposed-Med obtained macro- F_1 values of 0.8825, 0.8068, and 0.8403, whereas Proposed-Mean obtained 0.8742, 0.8030, and 0.8345. The corresponding values were 0.7654, 0.6940, and 0.7256 for BLINKER-concat and 0.6664, 0.5627, and 0.6086 for MNE-annot. This separation resulted from the precision–recall profiles shown in Figure 6: BLINKER-concat produced the highest recall of any condition on Raja, Cao2018, and the pooled evaluation, at 0.9518, 0.9715, and 0.9627, but also the lowest precision, at 0.6794, 0.5696, and 0.6182. Wilcoxon signed-rank tests on session-level F_1 , Bonferroni-corrected over the six pairwise comparisons, separated each proposed configuration from each baseline: Proposed-Med versus BLINKER-concat gave $p = 1.32 \times 10^{-11}$ and versus MNE-annot $p = 6.51 \times 10^{-9}$, while Proposed-Mean versus BLINKER-concat gave $p = 4.97 \times 10^{-10}$ and versus MNE-annot $p = 9.94 \times 10^{-9}$. Neither within-pair comparison was significant: BLINKER-concat versus MNE-annot yielded $p = 0.812$, and Proposed-Mean versus Proposed-Med yielded $p = 1$. No channel group on either dataset produced an inversion in which a baseline matched or exceeded Proposed-Med (Table 7).

1) Effect of the Threshold Estimator at Stage B

a: Motivation.

Stage B estimates the sample-level threshold from samples pooled across the suspicious epoch set $\mathcal{S}$. Because $\mathcal{S}$ is se-

lected precisely because it contains high-amplitude excursions, the pooled distribution is not symmetric around a clean baseline: it is contaminated by the very peaks the threshold is meant to capture. The mean and standard deviation are sensitive to those extreme values; the median and MAD are

b: Hypothesis.

Proposed-Med will outperform Proposed-Mean on precision—because the MAD-based threshold is less inflated by within-epoch peaks—while achieving comparable or superior recall.

Within the two proposed configurations, the Stage-B threshold estimators produced similar overall performance, although the direction of the metric changes did not support the stated precision hypothesis (Table 6). On Raja, Proposed-Med achieved precision, recall, and F_1 of 0.8803, 0.9014, and 0.8825, compared with 0.8840, 0.8876, and 0.8742 for Proposed-Mean. The corresponding changes were -0.0038 in precision, $+0.0138$ in recall, and $+0.0083$ in F_1 . On Cao2018, Proposed-Med likewise reduced precision by -0.0116 while increasing recall by $+0.0190$ and F_1 by $+0.0037$. Pooled performance followed the same pattern, with precision decreasing from 0.8209 to 0.8127, recall increasing from 0.8789 to 0.8956, and F_1 increasing from 0.8345 to 0.8403. However, the two estimators were statistically indistinguishable under the two-sided Bonferroni-corrected Wilcoxon test ($p = 1$). Proposed-Med exceeded Proposed-Mean in 46/104 sessions, was equal in 4, and was lower in 54, indicating a per-session outcome close to even.

2) Cross-Dataset Generalisation

a: Motivation.

A detector tuned and reported on a single corpus may not transfer to data collected under different acquisition conditions. Because the benchmark spans a closed corpus (Raja) and an open corpus (Cao2018), the generalisation gap can be quantified directly as the per-condition difference in macro- F_1 between the two datasets.

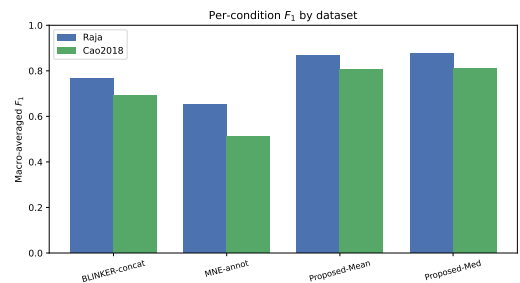


FIGURE 7. Macro-averaged F_1 per condition on Raja and Cao2018.

Across datasets, macro F_1 was consistently higher on Raja than on Cao2018 under every condition (Table 8; Figure 7). BLINKER-concat achieved 0.7654 on Raja and 0.6940 on Cao2018, corresponding to a cross-dataset gap of $+0.0714$. MNE-annot achieved 0.6664 and 0.5627, respectively, with

TABLE 6. Strategy comparison on the Raja and Cao2018 driving-EEG corpora at 30 s epochs. Each condition is summarised at its best-channel-per-session operating point (for every session the single channel or frontal sub-montage with the highest F_1 is selected, then averaged over sessions); the same rule is applied to all four conditions. Macro-averaged precision, recall and F_1 are reported per dataset and pooled over all 104 sessions. Best F_1 per block in bold. By paired Wilcoxon signed-rank tests on session-level F_1 (Bonferroni-corrected over six pairs), Proposed-Med significantly exceeds BLINKER-concat ($p < 10^{-9}$) and MNE-annot ($p < 0.001$) but not Proposed-Mean ($p = 1.00$).

Dataset	Condition	Precision	Recall	F_1
Raja	BLINKER-concat	0.6794	0.9518	0.7654
	MNE-annot	0.6944	0.7063	0.6664
	Proposed-Mean	0.8840	0.8876	0.8742
	Proposed-Med	0.8803	0.9014	0.8825
Cao2018	BLINKER-concat	0.5696	0.9715	0.6940
	MNE-annot	0.6013	0.6168	0.5627
	Proposed-Mean	0.7707	0.8720	0.8030
	Proposed-Med	0.7591	0.8909	0.8068
Pooled	BLINKER-concat	0.6182	0.9627	0.7256
	MNE-annot	0.6425	0.6564	0.6086
	Proposed-Mean	0.8209	0.8789	0.8345
	Proposed-Med	0.8127	0.8956	0.8403

TABLE 7. Channel groups and datasets where a baseline algorithm equals or exceeds Proposed-Med in best-channel-per-session macro- F_1 .

Dataset	Selection	Baseline	BL- F_1	Prop- F_1	ΔF_1
No inversions — Proposed-Med leads on every channel group.					

TABLE 8. Cross-dataset generalisation gap (best-channel-per-session macro- F_1 , 30 s epochs). Gap = Raja – Cao2018.

Condition	Raja	Cao2018	Gap
BLINKER-concat	0.7654	0.6940	+0.0714
MNE-annot	0.6664	0.5627	+0.1037
Proposed-Mean	0.8742	0.8030	+0.0712
Proposed-Med	0.8825	0.8068	+0.0758

the largest gap of +0.1037. Proposed-Mean achieved 0.8742 on Raja and 0.8030 on Cao2018, producing the smallest gap of +0.0712, whereas Proposed-Med achieved 0.8825 and 0.8068, respectively, with a gap of +0.0758. The two proposed configurations and BLINKER-concat therefore maintained similar, narrow differences between the two corpora, while MNE-annot showed the least consistent performance across datasets.

3) Operating Points

Across the pooled and corpus-specific analyses, the precision–recall operating points showed a consistent ordering among the proposed and baseline conditions (Figure 8 and Figure 9). In the pooled analysis, Proposed-Med and Proposed-Mean occupied the high-precision, high-recall region, with precision and recall of 0.8127 and 0.8956 for Proposed-Med and 0.8209 and 0.8789 for Proposed-Mean, respectively. BLINKER-concat occupied the high-recall, low-precision region, with recall of 0.9627 and precision of 0.6182, whereas MNE-annot was lower on both measures, with precision of 0.6425 and recall of 0.6564. The same ordering was observed within each corpus. On

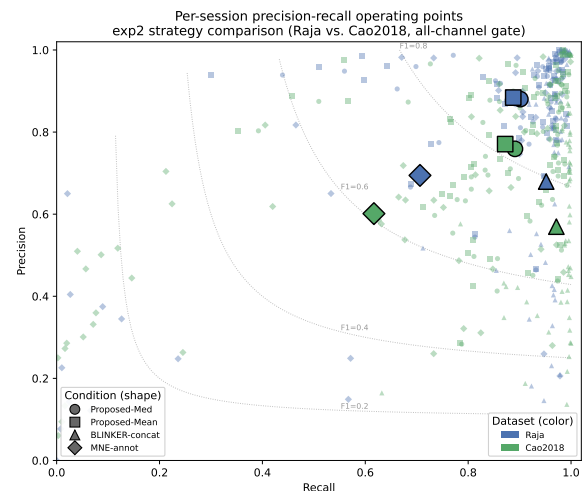


FIGURE 8. Per-session precision–recall operating points, Raja and Cao2018 shown separately.

Raja, the proposed conditions had precision of 0.8840 and 0.8803 and recall of 0.8876 and 0.9014, whereas BLINKER-concat combined 0.9518 recall with 0.6794 precision and MNE-annot achieved 0.6944 precision and 0.7063 recall. On Cao2018, the corresponding pattern remained, with the proposed conditions retaining higher precision and recall than MNE-annot, while BLINKER-concat retained the highest recall and lower precision.

4) Blink-Count Fidelity

a: Motivation.

Event-level F_1 does not by itself establish that a detector can support blink-rate or PERCLOS estimation: that requires the predicted event count to track the true count. For each session the predicted count is $TP + FP$ and the true count is $TP + FN$.

Beyond the precision–recall operating points, blink-count fidelity also differed across conditions (Table 9; Figure 10). BLINKER-concat produced a mean predicted-to-true count

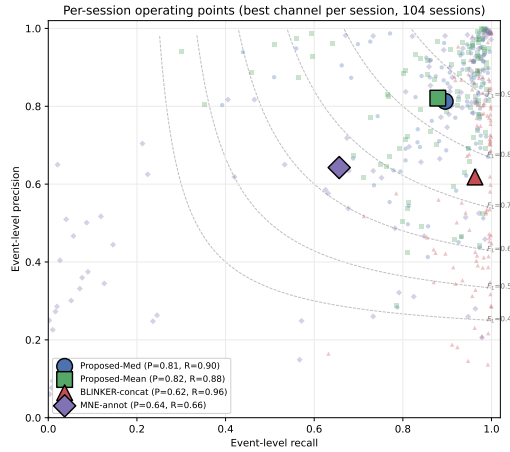


FIGURE 9. Per-session operating points pooled over both corpora, with condition means and iso- F_1 contours.

TABLE 9. Agreement between the predicted blink count (TP + FP) and the true count (TP + FN) per session, pooled over Raja and Cao2018 ($n = 104$) at the best-channel-per-session row. The mean count ratio measures systematic over- or under-counting (1.00 is ideal); Pearson r and Lin's concordance correlation coefficient (CCC) measure how closely the predicted count tracks the true count across sessions.

Condition	Mean count ratio	Pearson r	Lin's CCC
Proposed-Med	1.16	0.941	0.938
Proposed-Mean	1.12	0.918	0.917
BLINKER-concat	1.94	0.868	0.714
MNE-annot	1.04	0.629	0.617

ratio of 1.939, with Pearson $r = 0.868$ and Lin's concordance correlation coefficient of 0.714, indicating that it roughly doubled the true blink count despite retaining a positive association with the true counts. MNE-annot produced a count ratio close to one at 1.036, but its Pearson correlation and concordance were lower at $r = 0.629$ and 0.617, respectively. Its apparently accurate pooled total therefore resulted from offsetting over-counting and under-counting errors rather than consistent tracking of the true count. Proposed-Mean achieved a count ratio of 1.124, Pearson $r = 0.918$, and concordance of 0.917, whereas Proposed-Med achieved 1.157, $r = 0.941$, and 0.938, respectively. For Proposed-Med, Bland-Altman analysis gave a bias of +80.2 events, with $SD=303.1$ and 95% limits of agreement from -513.9 to $+674.4$, showing that it still over-counted on average.

5) Error-Structure Decomposition

a: Motivation.

Two detectors can reach the same F_1 by opposite routes: one floods the recording with false positives, the other misses blinks. Decomposing each condition's errors into false-positive and false-negative regimes distinguishes the two.

The corresponding error profiles differed across conditions (Table 10), particularly in the balance between false positives and false negatives. BLINKER-concat produced 627.9

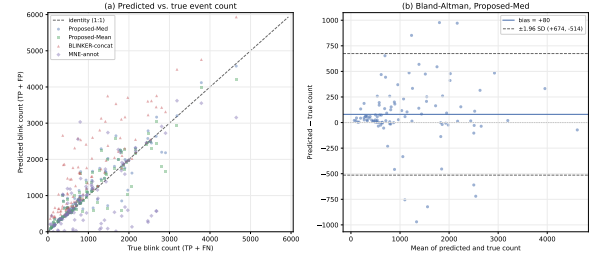


FIGURE 10. Predicted versus true blink count, and a Bland-Altman view for Proposed-Med.

TABLE 10. Error-structure decomposition by condition (best-channel-per-session). Mean false positives (FP) and false negatives (FN) per session, pooled over 104 sessions. The regime column records which error type dominates.

Condition	Mean FP/session	Mean FN/session	FP:FN	Regime
BLINKER-concat	627.9	23.5	26.675	FP-heavy
MNE-annot	276.7	451.5	0.613	FN-heavy
Proposed-Mean	191.3	158.7	1.206	FP-heavy
Proposed-Med	208.0	127.7	1.628	FP-heavy

false positives and 23.5 false negatives per session, giving an FP:FN ratio of 26.675 and an overwhelmingly false-positive-heavy error profile. MNE-annot produced 276.7 false positives and 451.5 false negatives per session, with an FP:FN ratio of 0.613, indicating that its errors were false-negative-heavy. Both proposed configurations were closer to parity. Proposed-Mean produced 191.3 false positives and 158.7 false negatives per session, corresponding to an FP:FN ratio of 1.206, whereas Proposed-Med produced 208.0 false positives and 127.7 false negatives per session, with an FP:FN ratio of 1.628. Thus, both proposed configurations remained moderately false-positive-heavy, but Proposed-Med traded additional false positives for fewer false negatives than Proposed-Mean.

6) Per-Session Variability and Failure Analysis

TABLE 11. Best and worst Proposed-Med sessions and subject-level summary across 104 Raja+Cao2018 sessions (best-channel-per-session).

Scope	Dataset	Unit	n	Metric
Best session	Raja	S21/S34_20190122_044130_3	1	F_1
Worst session	Cao2018	S53/090925m	1	F_1
Median session	all	104 sessions	104	F_1
Best subject	Cao2018	S55	1	Mean F_1
Worst subject	Cao2018	S31	2	Mean F_1
Median subject	all	44 subjects	44	Mean F_1

Beyond these condition-specific error profiles, Proposed-Med showed substantial variability across sessions and subjects, and the lowest-scoring sessions did not share a uniform failure pattern (Table 11). Across 104 sessions, session-level F_1 ranged from 0.9944 for S21/S34_20190122_044130_3 in Raja to 0.4167 for S53/090925m in Cao2018, with a median of 0.8730 and an interquartile range of [0.7659,

TABLE 12. The 5 lowest- F_1 Proposed-Med sessions per corpus (best channel per session). GT is the ground-truth blink count (TP + FN) and GT/med is GT relative to the dataset median, which separates genuine detector failure from recordings that simply carry an atypical number of blinks.

Dataset	Session	GT	TP	FP	FN	Regime	F_1	GT/med	Best channel
Raja	S16/S29_20190111_034326_3	1822	800	52	1022	FN-heavy	0.598	$3.5\times$	Fp2
	S1/S01_20170519_043933_3	132	123	142	9	FP-heavy	0.620	$0.3\times$	AF4
	S3/S03_20170605_033654_3	91	74	62	17	FP-heavy	0.652	$0.2\times$	Fp1
	S18/S31_20190115_035853_2	455	313	157	142	FP-heavy	0.677	$0.9\times$	Fp1
	S24/S38_20190129_035118_2	1117	625	34	492	FN-heavy	0.704	$2.1\times$	Fp2
Cao2018	S53/090925m	367	289	731	78	FP-heavy	0.417	$0.3\times$	Fp1
	S31/061020m	1476	579	142	897	FN-heavy	0.527	$1.1\times$	Fp1
	S49/080527n	174	149	212	25	FP-heavy	0.557	$0.1\times$	F7
	S22/090825n	382	372	479	10	FP-heavy	0.603	$0.3\times$	Fp1
	S53/090918n	809	749	912	60	FP-heavy	0.606	$0.6\times$	Fp1

0.9360]. Across 44 subjects, mean F_1 ranged from 0.9928 for S55 to 0.5856 for S31, with a median subject mean of 0.8350. In Raja, 1 of 46 sessions had $F_1 < 0.6$ and 4 had $F_1 < 0.7$, whereas in Cao2018, 3 of 58 sessions had $F_1 < 0.6$ and 12 had $F_1 < 0.7$ (Table 12). The lowest-scoring sessions were not predominantly associated with anomalously high ground-truth blink counts: the bottom-5 median was $0.9\times$ the Raja dataset median and $0.3\times$ the Cao2018 dataset median. The bottom groups were also not uniformly false-negative-dominated, with FN-heavy cases in 2/5 Raja sessions and 1/5 Cao2018 sessions.

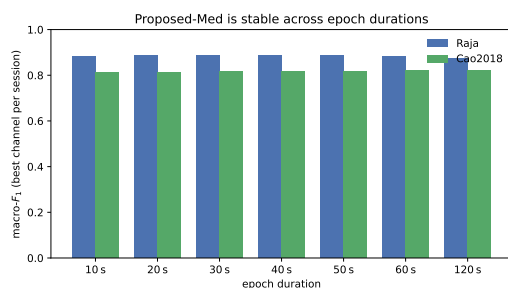
E. EXPERIMENT 3: STABILITY ACROSS EPOCH DURATIONS

a: Motivation.

Epoch duration is routinely chosen on paradigm-specific or theoretical grounds and is rarely optimised for the blink detector itself. In driving studies, epoch lengths ranging from a few seconds to several minutes appear in the literature. A robust pipeline should produce stable event-level F_1 across this range, because the underlying physiology of blinks does not change with the epoch grid imposed during analysis.

b: Design.

Proposed-Med is evaluated under epoch durations of 10, 20, 30, 40, 50, 60 and 120 seconds. For each duration the full pipeline is re-run from scratch, since epoch-level peak-to-peak statistics change with epoch length. Table 13 reports results and two-tailed Wilcoxon p -values relative to the 30-second reference.

**FIGURE 11.** Macro-averaged F_1 of Proposed-Med across epoch durations.**TABLE 13.** Best-channel-per-session macro- F_1 of Proposed-Med across epoch durations. p -values (two-tailed Wilcoxon on session-level F_1 , Bonferroni-corrected over 6 non-reference durations) compare each duration against the 30 s reference. Bold marks the best duration within each block.

Dataset	Epoch duration	n	Macro F_1	p vs. 30 s
Raja	10 s	46	0.8842	1.000
	20 s	46	0.8879	1.000
	30 s	46	0.8882	reference
	40 s	46	0.8867	0.884
	50 s	46	0.8860	0.112
	60 s	46	0.8837	0.041
	120 s	46	0.8720	< 0.001
Cao2018	10 s	58	0.8136	1.000
	20 s	58	0.8140	1.000
	30 s	58	0.8153	reference
	40 s	58	0.8150	1.000
	50 s	58	0.8179	1.000
	60 s	58	0.8207	0.109
	120 s	58	0.8200	0.197
Pooled	10 s	104	0.8448	1.000
	20 s	104	0.8467	0.497
	30 s	104	0.8476	reference
	40 s	104	0.8467	1.000
	50 s	104	0.8480	1.000
	60 s	104	0.8486	1.000
	120 s	104	0.8430	0.097

Epoch-duration analysis indicated that pooled macro- F_1 was stable overall, with Raja providing the only duration-specific exception at longer epochs (Table 13; Figure 11). Across the complete 10–120 s range, pooled macro- F_1 varied by 0.0056, from 0.8430 at 120 s to 0.8486 at 60 s, compared with 0.8476 at the 30 s reference. Pooled performance was statistically indistinguishable from the 30 s reference at every tested duration, with adjusted p -values ranging from 0.097 to 1.000. Cao2018 similarly showed a spread of 0.0071, with no duration differing significantly from the reference. In contrast, Raja showed a wider spread of 0.0163: macro- F_1 fell from 0.8882 at 30 s to 0.8720 at 120 s ($p < 0.001$), while the value at 60 s was also lower at 0.8837 ($p = 0.041$). Performance was therefore stable across 10–60 s overall. The 30 s duration was retained as the primary configuration for consistency with common practice in driving-fatigue EEG analysis rather than because it maximised macro- F_1 .

F. SUMMARY ACROSS EXPERIMENTS

Table 14 consolidates Proposed-Med performance across the three experiments, Figure 12 shows the corresponding session-level distributions, and Table 15 reports the paired comparison against the best competing method in every setting.

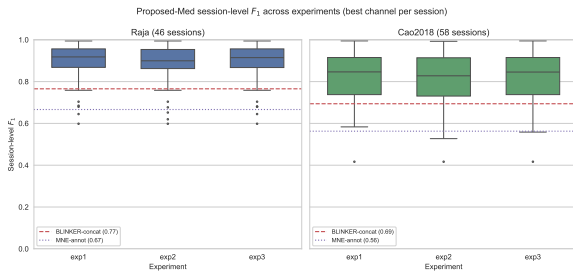


FIGURE 12. Session-level F_1 distributions of Proposed-Med across the three experiments.

Taken together, the three experiments showed that Proposed-Med consistently outperformed the best competing method, BLINKER-concat, on both Raja and Cao2018 (Table 14; Figure 12). At each experiment's primary configuration, pooled macro F_1 was 0.8493 for channel selection, 0.8403 for strategy comparison, and 0.8476 for the 30 s epoch duration, compared with 0.7256 for BLINKER-concat. The corresponding pooled mean advantages ranged from +0.1147 to +0.1237. The corpus-specific results followed the same ordering, with Proposed-Med achieving 0.8894, 0.8825, and 0.8882 on Raja and 0.8175, 0.8068, and 0.8153 on Cao2018 across Experiments 1–3, respectively. In the paired analyses, every bootstrap 95% confidence interval excluded zero, and all six one-sided Wilcoxon comparisons remained significant after Bonferroni correction (Table 15). Thus, the advantage of Proposed-Med was consistent across experimental settings and corpora.

IV. DISCUSSION

Deriving the amplitude threshold from epoch-screened signal rather than from the whole concatenated recording provides a substantial and statistically significant improvement in blink-region detection across independent driving corpora. Proposed-Med achieved a pooled macro- F_1 of 0.8403, compared with 0.7256 for BLINKER-concat and 0.6086 for MNE-annot (Table 6). BLINKER-concat represented an established blink-specific method based on thresholding the concatenated recording [8], whereas MNE-annot represented a community preprocessing baseline related to automated rejection and standardized preprocessing procedures [29, 30]. The corresponding session-level differences were statistically significant for both BLINKER-concat ($p = 1.32 \times 10^{-11}$) and MNE-annot ($p = 6.51 \times 10^{-9}$), and neither baseline equalled or exceeded Proposed-Med in any channel group on either dataset. These findings indicate that estimating the threshold after excluding unsuitable epochs reduces the influence of recording-wide amplitude heterogeneity and yields

more consistent region-level discrimination than applying a single threshold derived from the complete signal. This improvement is consistent with reduced contamination of the reference amplitude distribution by heterogeneous epoch composition. When blink-heavy and blink-free epochs were pooled across the concatenated recording, the large proportion of blink-free signal pulled the estimated threshold toward the baseline amplitude, causing the detector to fire too readily. The resulting error structure provides direct evidence for this mechanism: BLINKER-concat retained the highest pooled recall of 0.9627 but had a precision of 0.6182, with 627.9 false positives and 23.5 false negatives per session and an FP:FN ratio of 26.675. Its errors were therefore overwhelmingly false-positive-heavy rather than arising from missed blinks. This sensitivity is expected because ocular-artifact amplitude varies across subjects, electrodes, and recording sessions [6, 8, 11, 24]. A single threshold derived from the full recording consequently depends on the relative contribution of blink-rich and blink-poor periods, whereas epoch screening restricts threshold estimation to signal segments that better represent the local background. The observed precision gain therefore reflects improved calibration to session-specific amplitude structure rather than reduced sensitivity to true blinks. The pipeline's advantage therefore arises from screening epochs before threshold estimation [31] rather than from whether the retained amplitudes are centred by the median or the mean. The median/MAD centre was selected to reduce the influence of within-epoch peaks that remained in the suspicious-epoch pool, consistent with the general robustness of median-based estimators to outlying observations [32, 33]. This rationale was not supported by a measurable performance gain in the present benchmark. Relative to Proposed-Mean, Proposed-Med reduced pooled precision by 0.0081 while increasing recall by 0.0167, yielding only a 0.0057 increase in F_1 . The two estimators were statistically indistinguishable after correction ($p = 1$), and Proposed-Med exceeded Proposed-Mean in 46/104 sessions, with 4 equal and 54 lower. Thus, the median centre did not provide the hypothesised precision benefit and won only about half of the sessions. Its resistance to contaminated amplitude samples makes it a defensible default on statistical grounds, but the present results do not support it as a source of measurable improvement.

A. SPATIAL REQUIREMENTS

The spatial requirements of the pipeline were limited because a single frontopolar electrode recovered essentially all of the whole-cap oracle performance in both corpora. On Raja, Fp2 achieved a macro- F_1 of 0.868 compared with the oracle value of 0.8894, whereas on Cao2018, Fp1 achieved 0.797 compared with 0.8175. The optimum therefore localised consistently to Fp1/Fp2 despite differences in montage and acquisition hardware, while performance declined at electrodes farther from the eyes. This spatial consistency is compatible with the strong projection of blink activity to the prefrontal scalp and indicates that additional channels contributed little to blink-region detection once a frontopolar signal was

TABLE 14. Proposed-Med detection performance across the three experiments on the Raja and Cao2018 driving-EEG corpora. Macro-averaged F_1 over all sessions (best-channel-per-session). The best competing method is BLINKER-concat from the strategy comparison; Δ is the pooled Proposed-Med advantage over BLINKER-concat (pooled macro- F_1 0.7256).

Exp.	Description	PM F_1 (Raja)	PM F_1 (Cao2018)	Best competitor	Δ vs competitor
exp1	Channel selection (whole-cap oracle)	0.8894	0.8175	BLINKER-concat	+0.1237
exp2	Strategy comparison	0.8825	0.8068	BLINKER-concat	+0.1147
exp3	Epoch duration (30 s)	0.8882	0.8153	BLINKER-concat	+0.1220

TABLE 15. Paired Wilcoxon signed-rank tests of Proposed-Med versus the best competing method (BLINKER-concat from the strategy comparison) on session-level F_1 , for each experiment and dataset. ΔF_1 is the mean Proposed-Med advantage; p is Bonferroni-corrected over the 6 comparisons (one-sided, Proposed-Med greater); r is the rank-biserial effect size; the 95% CI is a 10,000-sample bootstrap on ΔF_1 .

Exp.	Dataset	ΔF_1	W	p_{Bonf}	r	95% CI	n
exp1	Raja	+0.1239	911	$< 10^{-5}$	0.157	[+0.0715, +0.1787]	46
exp1	Cao2018	+0.1235	1647	$< 10^{-9}$	0.037	[+0.0966, +0.1502]	58
exp2	Raja	+0.1171	909	$< 10^{-5}$	0.159	[+0.0655, +0.1699]	46
exp2	Cao2018	+0.1127	1613	$< 10^{-8}$	0.057	[+0.0860, +0.1398]	58
exp3	Raja	+0.1228	911	$< 10^{-5}$	0.157	[+0.0693, +0.1761]	46
exp3	Cao2018	+0.1213	1647	$< 10^{-9}$	0.037	[+0.0943, +0.1481]	58

available. The result also agrees with prior single-channel and forehead-EEG studies that detected blinks or derived fatigue-related measures from prefrontal recordings [10, 12, 15, 16, 21, 23, 34–36]. A dense montage is therefore not required for this task, which supports implementations based on a single dry or embedded forehead electrode in wearable and in-vehicle systems where setup time, comfort, and sensor count constrain deployment. Within this reduced spatial requirement, channel selection was stable at the level of scalp region but not at the level of a single electrode. Fp1 and Fp2 accounted for the large majority of selections in both corpora: Raja favoured Fp2 in 75/184 selections and Fp1 in 67/184, whereas Cao2018 favoured Fp1 in 119/232 and Fp2 in 57/232. However, the four conditions selected the same channel in only 9/46 Raja sessions and 6/58 Cao2018 sessions, with pooled full agreement of 15/104. This apparent tension indicates that the detector consistently localised blink information to the frontopolar region while the electrode carrying the strongest signal varied across conditions and sessions. Such variation is consistent with lateralised ocular projections and inter-subject differences in blink topography, electrode placement, and signal amplitude rather than with instability of the underlying detection principle [24, 37]. The practical implication is that deployment should retain both frontopolar sites and select between them adaptively, rather than fixing acquisition or detection to Fp1 or Fp2 alone.

B. GENERALISATION AND ROBUSTNESS

Beyond this spatial consideration, cross-dataset generalisation was configuration-dependent: all conditions achieved higher macro- F_1 on Raja than on Cao2018, but the magnitude of this difference varied by configuration. Proposed-Mean and Proposed-Med showed gaps of +0.0712 and +0.0758, respectively, which were close to the +0.0714 gap obtained by BLINKER-concat. By contrast, MNE-annot showed the largest gap at +0.1037 and was therefore the least consistent

across the two corpora. This pattern indicates that the lower scores on Cao2018 are more plausibly attributable to differences between the corpora than to overfitting specific to the proposed thresholding procedure. Cao2018 was collected as an open multichannel sustained-attention driving dataset [25], and differences in acquisition hardware, electrode montage, and recording conditions can alter blink amplitude and morphology across datasets. The proposed configurations nevertheless retained a cross-dataset gap comparable to that of the established blink-specific baseline [8]. Their advantage was therefore not confined to Raja, although absolute performance remained dependent on the recording corpus. In contrast to the configuration-dependent cross-dataset differences, epoch-duration sensitivity was limited overall, although the longest tested epoch produced a corpus-dependent decline. Pooled performance was statistically indistinguishable from the 30 s reference at every tested duration from 10 to 120 s, with macro- F_1 varying only from 0.8430 to 0.8486. This stability matters because epoch duration in driving studies is usually determined by the experimental paradigm rather than by the blink detector, and the thresholding procedure therefore needs to tolerate different segmentation choices. The exception occurred on Raja, where performance at 120 s fell from 0.8882 at the 30 s reference to 0.8720 and was significantly lower ($p = 0.000$). This decline is consistent with dilution of the screening stage: as an epoch becomes longer, it is more likely to contain both blink and blink-free signal, so its summary amplitude becomes less representative of either state. That mechanism is the same whole-recording dilution that the proposed pipeline was designed to avoid by estimating the threshold from screened epochs. The absence of a corresponding pooled decline indicates that the effect was corpus-dependent rather than a general duration sensitivity, but the Raja result argues against treating very long epochs as interchangeable with shorter segments. Epochs of 10–60 s therefore provide the safer operating range.

C. DOWNSTREAM USABILITY

For downstream usability, event-level F_1 did not by itself establish suitability for blink-rate or PERCLOS-style measures, because these applications require predicted counts to vary with the true counts across recordings [2, 4, 38, 39]. Proposed-Med provided the closest count fidelity, with a mean predicted-to-true ratio of 1.157, Pearson $r = 0.941$, and Lin CCC = 0.938. BLINKER-concat instead produced approximately twice the true count, with a ratio of 1.939, despite retaining a relatively high correlation of $r = 0.868$; this pattern indicates systematic inflation rather than accurate absolute measurement. MNE-annot showed the converse limitation: its ratio of 1.036 was near unity, but its lower correlation of $r = 0.629$ and CCC = 0.617 indicate that over-counting and under-counting offset in the pooled mean rather than that session-level counts were faithfully tracked. Proposed-Med therefore better preserves between-session variation needed for downstream ocular indices. However, its positive bias of +80.2 events and ratio above unity show that it still over-counted, so absolute blink rates and derived thresholds should be interpreted with care. Consistent with this downstream focus, the error profiles placed the two baselines at opposite operating-point extremes, whereas the proposed methods provided a closer balance between false positives and false negatives. BLINKER-concat was strongly false-positive dominated, with an FP:FN ratio of 26.675, whereas MNE-annot was false-negative dominated, with a ratio of 0.613. By contrast, Proposed-Mean and Proposed-Med produced ratios of 1.206 and 1.628, respectively, indicating substantially closer correspondence between over-detection and missed events. This balance matters because methods with similar F_1 can impose different downstream costs: excessive false positives inflate estimated blink counts and increase the burden of rejecting spurious regions, whereas excessive false negatives suppress genuine events and bias analyses that depend on blink occurrence or timing. An operating point closer to parity therefore provides a more usable compromise when neither error type can be treated as negligible. Precision-recall characterisation is the appropriate framework for this comparison because blink detection is an imbalanced event-detection problem in which the positive class is sparse relative to continuous background signal [27]. Event-level evaluation is also preferable to sample-wise agreement because it assesses whether temporally localised detections correspond to distinct underlying events [28]. Beyond these aggregate error profiles, performance variability across sessions was not primarily explained by event density but was more consistent with differences in signal quality and the sensitivity of sparse-blink sessions to individual detection errors. Performance varied substantially across the 104 sessions, from an F_1 of 0.9944 to 0.4167, with a median of 0.8730 and an IQR of [0.7659, 0.9360]. Subject-level means likewise ranged from 0.9928 to 0.5856, with a median of 0.8350. However, the lowest-scoring sessions were not those with anomalously high blink counts. The bottom five sessions had median ground-truth counts of

0.9x the dataset median on Raja and only 0.3x on Cao2018, despite dataset medians of 526 and 1314, respectively. Their errors were also not uniformly dominated by missed events, because only 2/5 Raja sessions and 1/5 Cao2018 sessions were FN-heavy. This pattern argues against event density as the primary failure mechanism. In sparse-blink sessions, a small number of detection errors imposes a larger penalty on the per-session score. Ambiguity in blink definition and heterogeneity in artifact morphology may further increase this sensitivity across recordings [1, 40].

D. POSITIONING AGAINST PRIOR WORK

The proposed detector is positioned qualitatively rather than as numerically superior to earlier blink detectors because direct numerical ranking would be misleading given differences in datasets, blink definitions, sampling rates, and scoring procedures, as summarised in Table 16. In particular, matching criteria ranged from sample-level classification to event-level detection, so apparently similar performance measures did not necessarily represent the same detection requirement. This heterogeneity applies to threshold-based and unsupervised approaches alike [8, 9, 13, 15–17, 22]. Its performance was competitive with interpretable single-channel threshold detectors while being assessed under a strict event-level overlap criterion on both independent driving corpora. This evaluation required each predicted region to correspond adequately to a reference blink event rather than rewarding isolated correctly labelled samples. The detector also avoided the dataset-specific threshold tuning commonly used by comparable approaches, which indicates that its relative performance was obtained under a more transferable operating procedure rather than through adaptation to a particular recording set.

TABLE 16. Qualitative positioning of the present detector against prior single-channel and threshold-based blink/EOG detectors cited in this work. Because the prior studies use different datasets, blink definitions, sampling rates and especially different matching criteria (sample-level versus event-level), their reported metrics are not directly comparable to an event-level F_1 ; such cells are marked “n/r” (not reported in comparable form) rather than populated with non-comparable numbers. Only the final row reports quantitative detection figures, recomputed at the best-channel-per-session event-level operating point (macro- F_1 on Raja / Cao2018). The comparison is therefore qualitative by design.

Method	Signal channel	/	Evaluation data	Matching criterion	Reported metric
Chang et al. [15]	et Single pre-frontal EEG		Own EEG	n/r	n/r
Tran et al. [16]	Single/few-channel EEG		Own EEG	n/r	n/r
Zhang et al. [12]	et Frontal EEG (real-time)		Own EEG	n/r	n/r
Wang et al. [21]	et Single-channel EEG		Own EEG	n/r	n/r
BLINKER [8]	Single channel (EEG/EOG)		Multiple corpora	n/r	n/r
Cao et al. [22]	Multi-channel EEG		Own EEG	n/r	n/r
Agarwal & Sivakumar [9]	Single-channel EEG		Own EEG	n/r	n/r
Valderrama et al. [23]	et Single-channel EOG/EEG		Own recordings	n/r	n/r
Guttmann-Flury et al. [24]	et Multi-channel EEG		Own EEG	n/r	n/r
This work (Proposed-Med)	Single frontopolar EEG		Raja + Cao2018 driving	Event-level overlap (IoU 0.1)	macro- F_1 0.88 / 0.81

E. LIMITATIONS AND FUTURE WORK

Beyond the constraints on direct comparison with earlier detectors, this study had several limitations. The benchmark was restricted to both driving corpora, and generalisation to other experimental paradigms or clinical populations was therefore not tested. Event detection was evaluated using a single matching tolerance, although the reported scores would change under a stricter overlap criterion; absolute F_1 values should consequently not be compared across studies without accounting for differences in event-matching rules [28]. Recall was not stratified by blink duration, so performance on prolonged eye closures, which are particularly relevant to drowsiness assessment, was not established. Ground truth was based on expert annotation of frontopolar EEG and therefore inherited the annotator’s operational definition of blink onset and offset, which remains non-trivial because blink boundaries depend on the signal representation and adopted criteria [1]. Finally, the proposed method is a thresholding detector and was not evaluated as superior to learned detectors such as transformer-, vision-, and deep-learning-based approaches [41–43]. Its intended advantages are instead methodological simplicity, interpretability, and the ab-

sence of per-dataset training. Addressing these limitations requires future work on dedicated long-closure detection, broader benchmarking, and integration with learned detectors. A dedicated long-closure module should first score normal blinks and prolonged closures separately. This distinction is necessary because the events most directly associated with drowsiness are sustained eye closures rather than ordinary blinks, and aggregating both categories under a single detection criterion may obscure the signal of greatest operational relevance [3, 5]. The benchmark should also be extended beyond driving data to other epoch-structured paradigms, including settings in which task blocks, trials, or recording segments provide natural units for estimating a representative signal distribution. Such evaluation would determine whether epoch screening remains useful when blink morphology, cognitive demands, and recording conditions differ from those in sustained driving. A further direction is to combine the screening stage with learned blink and drowsiness detectors rather than treating the approaches as alternatives [44–46]. In that configuration, epoch screening would serve as a computationally inexpensive front end that limits implausible candidate regions before a learned model performs the more specific classification.

V. CONCLUSION

The study addressed blink-region detection in epoch-structured EEG, for which concatenated-signal methods pooled blink-containing and blink-free epochs when estimating detection thresholds. The proposed pipeline screened epochs by peak-to-peak amplitude, estimated a sample-level threshold from suspicious epochs using either the mean or median centre with MAD-based scaling, applied the threshold to extract blink regions, and mapped the detections to epoch-local coordinates. At 30 s, the proposed variants improved pooled macro-averaged F_1 from 0.7256 for BLINKER-concat and 0.6086 for MNE-annot to 0.8345 for Proposed-Mean and 0.8403 for Proposed-Med. This improvement chiefly reflected higher precision, with pooled values of 0.8209 and 0.8127 compared with 0.6182 and 0.6425 for the baselines, while recall remained 0.8789 and 0.8956. Proposed-Mean and Proposed-Med were statistically indistinguishable on session-level F_1 ($p = 1$), indicating that the advantage arose from epoch-aware screening rather than from the choice of centre.

The remaining experiments showed that the pipeline retained its performance under reduced channel coverage and changes in epoch duration. A single frontopolar electrode recovered most of the whole-cap oracle performance, with F_1 values of 0.868 for Fp2 compared with 0.8894 in Raja and 0.797 for Fp1 compared with 0.8175 in Cao2018. Pooled macro F_1 also remained stable from 10 s to 120 s, with a spread of 0.0056 and values ranging from 0.8430 to 0.8486. The predicted blink count tracked the true count closely for both proposed variants, with Pearson $r = 0.918$ and Lin’s CCC = 0.917 for Proposed-Mean and Pearson $r = 0.941$ and Lin’s CCC = 0.938 for Proposed-Med. This agreement

was sufficient to support blink-rate-style measures, although Proposed-Med retained a positive bias of +80.2 events and 95% limits of $[-513.9, +674.4]$.

The conclusions were limited by evaluation on the Raja and Cao2018 driving corpora, the use of a single event-matching tolerance, and the absence of stratification by blink duration. Future work will therefore assess the pipeline on additional recording settings and matching criteria, examine performance across blink-duration groups, and develop the planned dedicated long-closure module.

REFERENCES

- [1] M. Nyström, R. Andersson, D. Niehorster, R. Hessels, and I. Hooge, "What is a blink? classifying and characterizing blinks in eye openness signals," *Behavior Research Methods*, 2024, csv_key=28ULZ92A.
- [2] E. Alyan, "Blink-related eeg activity measures cognitive load during proactive and reactive driving," *Scientific Reports*, 2023, csv_key=alyan2023blink.
- [3] Y. Tian and J. Cao, "Fatigue driving detection based on electrooculography: a review," *Eurasip Journal on Image and Video Processing*, 2021, csv_key=GNNI8WH2.
- [4] P. Ren, X. Ma, W. Lai, M. Zhang, S. Liu, Y. Wang, M. Li, D. Ma, Y. Dong, Y. He, and X. Xu, "Comparison of the use of blink rate and blink rate variability for mental state recognition," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2019, csv_key=TUNW8X4S.
- [5] M. Shahbakhti, M. Beiramvand, E. Nasiri, S. Far, W. Chen, J. Sole-Casals, M. Wierzchon, A. Broniec-Wojcik, P. Augustyniak, and V. Marozas, "Fusion of eeg and eye blink analysis for detection of driver fatigue," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2023, csv_key=WL52M5AF.
- [6] R. J. Croft and R. J. Barry, "Removal of ocular artifact from the eeg: a review," *Neurophysiologie Clinique/Clinical Neurophysiology*, 2000, csv_key=croft2000removal.
- [7] S. Akuthota, K. Kumar, and R. Janapati, "Artifacts removal techniques in eeg data for bci applications: A survey," *Computational Intelligence and Deep Learning Methods for Neuro-rehabilitation Applications*, 2023, csv_key=7FIPNU96.
- [8] K. K. Kleifges, N. Bigdely-Shamlo, S. E. Kerick, and K. A. Robbins, "Blinker: Automated extraction of ocular indices from eeg enabling large-scale analysis," *Frontiers in Neuroscience*, 2017, csv_key=kleifges2017blinker.
- [9] M. Agarwal and R. Sivakumar, "Blink: A fully automated unsupervised algorithm for eye-blink detection in eeg signals," 2019, csv_key=C2X825UQ.
- [10] L.-W. Ko, O. Komarov, W.-K. Lai, W.-G. Liang, and T.-P. Jung, "Eyeblink recognition improves fatigue prediction from single-channel forehead eeg in a realistic sustained attention task," *Journal of Neural Engineering*, 2020, csv_key=Y3Y2KSSK.
- [11] A. Klein and W. Skrandies, "A reliable statistical method to detect eyeblink-artefacts from electroencephalogram data only," *Brain Topography*, 2013, csv_key=klein2013reliable.
- [12] Y. Zhang, X. Zheng, W. Xu, and H. Liu, "Rt-blink: A method toward real-time blink detection from single frontal eeg signal," *IEEE Sensors Journal*, 2023, csv_key=JFTJCTE7.
- [13] M. Rashida and M. Habib, "Quantitative eeg features and machine learning classifiers for eye-blink artifact detection: A comparative study," *Neuroscience Informatics*, 2023, csv_key=G28PH5JQ.
- [14] Y. Jiang, D. Wu, J. Cao, L. Jiang, S. Zhang, and D. Wang, "Eyeblink detection algorithm based on joint optimization of vme and morphological feature extraction," *IEEE Sensors Journal*, 2023, csv_key=QAZH DUHB.
- [15] W.-D. Chang, H.-S. Cha, K. Kim, and C.-H. Im, "Detection of eye blink artifacts from single prefrontal channel electroencephalogram," *Computer Methods and Programs in Biomedicine*, 2016, csv_key=EXMLDRXM.
- [16] D.-K. Tran, T.-H. Nguyen, and T.-N. Nguyen, "Detection of eeg-based eye-blinks using a thresholding algorithm," *European Journal of Engineering and Technology Research*, 2021, csv_key=tran2021detection.
- [17] L. Jefri, F. Rahman, N. Malik, and F. Isa, "Eye blink region detection in eeg using amplitude thresholding," 2022, csv_key=ECTDKUZE.
- [18] H. Nolan, R. Whelan, and R. Reilly, "Faster: Fully automated statistical thresholding for eeg artifact rejection," *Journal of Neuroscience Methods*, vol. 192, no. 1, pp. 152–162, 2010. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0165027010003894>
- [19] R. Ghosh, S. Phadikar, N. Deb, N. Sinha, P. Das, and E. Ghaderpour, "Automatic eyeblink and muscular artifact detection and removal from eeg signals using k-nearest neighbor classifier and long short-term memory networks," *IEEE Sensors Journal*, 2023, csv_key=D62QMUQY.
- [20] M. Wang, J. Wang, X. Cui, T. Wang, T. Jiang, F. Gao, and J. Cao, "Multidimensional feature optimization based eye blink detection under epileptiform discharges," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2022, csv_key=ARYZLKPY.
- [21] G. Wang, "Sliding window higher-order cumulants for detection of eye blink artifact from short segments of single-channel eeg," *PeerJ Computer Science*, 2025, csv_key=wang2025sliding.
- [22] J. Cao, L. Chen, D. Hu, F. Dong, T. Jiang, W. Gao, and F. Gao, "Unsupervised eye blink artifact detection from eeg with gaussian mixture model," *IEEE Journal of Biomedical and Health Informatics*, 2021, csv_key=H22KDJS.

- [23] J. Valderrama, A. De La Torre, and B. Van Dun, "An automatic algorithm for blink-artifact suppression based on iterative template matching: Application to single channel recording of cortical auditory evoked potentials," *Journal of Neural Engineering*, 2018, csv_key=HK732W3F.
- [24] E. Guttmann-Flury, X. Sheng, D. Zhang, and X. Zhu, "A new algorithm for blink correction adaptive to inter- and intra-subject variability," *Computers in Biology and Medicine*, 2019, csv_key=guttmann2019new.
- [25] Z. Cao, C.-H. Chuang, J.-K. King, and C.-T. Lin, "Multi-channel eeg recordings during a sustained-attention driving task," *Scientific Data*, 2019, csv_key=cao2019multichannel.
- [26] S. Dari, N. Eppele, and V. Protschky, "Unsupervised blink detection and driver drowsiness metrics on naturalistic driving data," 2020, csv_key=TZBBHJWH.
- [27] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets," *PLOS ONE*, 2015, csv_key=saito2015precision.
- [28] R. Salles, "Softed: Metrics for soft evaluation of time series event detection," *Computers & Industrial Engineering*, 2024, csv_key=salles2024softed.
- [29] M. Jas, D. A. Engemann, Y. Bekhti, F. Raimondo, and A. Gramfort, "Autoreject: Automated artifact rejection for MEG and EEG data," *NeuroImage*, vol. 159, pp. 417–429, 2017.
- [30] N. Bigdely-Shamlo, T. Mullen, C. Kothe, K.-M. Su, and K. A. Robbins, "The PREP pipeline: Standardized preprocessing for large-scale EEG analysis," *Frontiers in Neuroinformatics*, vol. 9, p. 16, 2015.
- [31] D.-K. Tran, T.-H. Nguyen, and B.-V. Ngo, "Amplitude thresholding of eeg signals for eye blink and saccade detection," 2021, pp. 268–273.
- [32] C. Leys, C. Ley, O. Klein, P. Bernard, and L. Licata, "Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median," *Journal of Experimental Social Psychology*, vol. 49, no. 4, pp. 764–766, 2013.
- [33] P. J. Rousseeuw and C. Croux, "Alternatives to the median absolute deviation," *Journal of the American Statistical Association*, vol. 88, no. 424, pp. 1273–1283, 1993.
- [34] M. Shahbakhti, M. Beiramvand, I. Rejer, P. Augustyniak, A. Broniec-Wojcik, M. Wierchcon, and V. Marozas, "Simultaneous eye blink characterization and elimination from low-channel prefrontal eeg signals enhances driver drowsiness detection," *IEEE Journal of Biomedical and Health Informatics*, 2022, csv_key=KPMMW2C5.
- [35] Y. Aoki and T. Fukami, "Drowsiness detection based on eeg and blink information measured with a single electrode," *International Journal of Innovative Computing, Information and Control*, vol. 19, no. 1, pp. 229–238, 2023.
- [36] R. López-Ahumada, R. Jiménez-Naharro, and F. Gómez-Bravo, "A hardware-based configurable algorithm for eye blink signal detection using a single-channel bci headset," *Sensors*, vol. 23, no. 11, 2023.
- [37] M. Plöchl, J. P. Ossandón, and P. König, "Combining EEG and eye tracking: Identification, characterization, and correction of eye movement artifacts in electroencephalographic data," *Frontiers in Human Neuroscience*, vol. 6, p. 278, 2012.
- [38] L. Dai, Y. Li, and M. Zhang, "Detection of operator fatigue in the main control room of a nuclear power plant based on eye blink rate, perclos and mouse velocity," *Applied Sciences (Switzerland)*, 2023, csv_key=5J3QBV3S.
- [39] M. Zulkarnanie, K. Shanmugam, N. Badruddin, and M. Saad, "Enhancements to perclos algorithm for determining eye closures," 2022, csv_key=6C9D6CVR.
- [40] P. Arpaia, M. De Luca, L. Di Marino, D. Duran, L. Gargiulo, P. Lanteri, N. Moccaldi, M. Nalin, M. Picciafuoco, R. Robbio, and E. Visani, "A systematic review of techniques for artifact detection and artifact category identification in electroencephalography from wearable devices," *Sensors*, 2025, csv_key=LKNZDL8L.
- [41] Á. Fodor, K. Fenech, and A. Lőrincz, "Blinklinmult: Transformer-based eye blink detection," *Journal of Imaging*, 2023, csv_key=8LGNBAX.
- [42] J. Hong, J. Shin, J. Choi, and M. Ko, "Robust eye blink detection using dual embedding video vision transformer," 2024, csv_key=7IZFRWNY.
- [43] A. Egambaram and N. Badruddin, "An investigation to detect driver drowsiness from eye blink artifacts using deep learning models," 2022, pp. 23–29.
- [44] N. Cabot, D. Lamouchi, Y. Yaddaden, and R. Cherif, "Drowsiness detection through yawning and eye blinking models using convolutional neural networks and transfer learning," 2024, csv_key=WF8X73VC.
- [45] F. Makhmudov, D. Turimov, M. Xamidov, F. Nazarov, and Y.-I. Cho, "Real-time fatigue detection algorithms using machine learning for yawning and eye state," *Sensors*, 2024, csv_key=MMQCZ4P7.
- [46] B. Zhang, D. Yan, X. Wang, L. Wang, and J. Mo, "Fatigue driving detection based on fusion of eeg and blink potential features," *Journal of Railway Science and Engineering*, vol. 22, no. 8, pp. 3738–3747, 2025.